

Use-Case Flow Specification

Problem Statement #47: Domain & SSL Certificate Expiry Alert System

UC-03: Generate Expiry Alert (includes UC-08 Send Notification; extended by UC-04 Escalate Unacknowledged Alert)

Use Case

UC-03 – Generate Expiry Alert

Primary Actor(s)

SysAdmin (recipient), Security Officer (recipient on escalation)

Related Use Cases

Includes: UC-08 Send Notification | Triggered by: UC-01 Run WHOIS Domain Audit, UC-02 Run SSL/TLS Certificate Audit | Extended by: UC-04 Escalate Unacknowledged Alert

Preconditions

1. The domain/SSL monitoring engine has completed a scheduled WHOIS or TLS handshake audit for at least one monitored asset.
2. The asset's computed expiry date falls within a configured alert threshold (30, 15, or 3 days).
3. The Notification Gateway (email/SMS/chat integration) is reachable and configured for the assigned SysAdmin.

Postconditions

1. A new alert record is created with a unique Alert ID, status "Open", severity, and threshold that triggered it.
2. A notification has been dispatched to the assigned SysAdmin via the Notification Gateway.
3. The alert is visible on the compliance dashboard until it is acknowledged or resolved.

Main Success Scenario

1. Audit engine (UC-01/UC-02) reports that a monitored domain or certificate will expire within a configured threshold (30, 15, or 3 days).
2. System creates a new alert record, assigning a unique Alert ID, timestamp, asset identifier, severity, and the SysAdmin responsible for that asset.
3. System determines the notification channel(s) configured for the responsible SysAdmin (email, SMS, chat).
4. System includes UC-08 Send Notification, passing the alert details and chosen channel(s).
5. Notification Gateway delivers the message to the SysAdmin.
6. System logs the delivery confirmation and sets the alert status to "Open – Notified".
7. Alert appears on the compliance dashboard, visible to SysAdmin and Security Officer.
8. Use case ends successfully.

Alternate Flow – 6a. Notification Delivery Fails

- 6a1. Notification Gateway returns a delivery failure (e.g., invalid endpoint, service outage).
- 6a2. System retries delivery up to 2 additional times over the next 15 minutes.
- 6a3. If all retries fail, system marks the alert "Open – Delivery Failed", logs the failure reason, and immediately triggers UC-04 Escalate Unacknowledged Alert so the Security Officer is notified through a secondary channel.
- 6a4. Use case resumes at step 7 once the Security Officer channel confirms delivery.

Extension Point

If the alert remains unacknowledged 24 hours after step 7 (see FR-003), UC-04 Escalate Unacknowledged Alert extends this use case to notify the Security Officer.